SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

CSA3717 - SOFTWARE TESTING

EXPERIMENT 14

Online Examination System – Negative Test Case Study

Domain: Online Examination System

1. AIM

To identify and analyze the requirements of an Online Examination System and prepare negative test scenarios and test cases.

2. PROBLEM STATEMENT

The Online Examination System must reject wrong credentials, unauthorized exam access, attempts outside the scheduled time, invalid answer data, and late submissions.

3. SOFTWARE REQUIREMENT SPECIFICATION (SRS)

S.No.	Requirement
1	Wrong credentials must not grant access.
2	Exams cannot be accessed before their scheduled time.
3	Exams cannot accept submissions after the configured deadline except as explicitly designed.
4	Unauthorized exam URLs must be denied.
5	Invalid answer values must be rejected.
6	The timer and submission rules must be enforced.

4. DEFECT REPORT

Defect ID	Defect / Scenario	Expected Result	Actual Result	Severity	Status
NT01	Wrong login password	Valid ID and incorrect password	Login denied	Pass	
NT02	Access before schedule	Open exam before start time	Access prevented	Pass	
NT03	Access after deadline	Open exam after end time	Access prevented or closed	Pass	
NT04	Submit without required answers	Leave required answers blank	System warns or follows configured rule	Pass	
NT05	Unauthorized exam access	Open another student's exam URL	Access denied	Pass	
NT06	Submit after time expiry	Allow timer to expire and submit	Late submission must be prevented/handled correctly	Fail	
NT07	Invalid answer request	Send invalid option value	Invalid answer rejected	Pass	
NT08	Refresh during exam	Refresh active exam page	Saved answers/session handled safely	Pass	

5. RESULT

The Online Examination System requirements were analyzed and the negative test cases were prepared.

5. DETAILED EXECUTION OF A FAILED TEST CASE

Test Case NT06
Failed

Test scenario: Late submission after examination time

Precondition

A student is in an active examination and the configured end time has been reached.

Test Steps

1. Login.
2. Start scheduled exam.
3. Allow timer to reach zero.
4. Click Submit.
5. Observe submission status.

Expected Result

The system should close or auto-submit according to configured rules and must not accept a late submission.

Actual Result

The system accepts the late submission after the examination time has expired.

Conclusion

Examination time restrictions are not enforced correctly.

6. SAMPLE BUG REPORT

Field	Details
Bug ID	BR-001
Linked Test Case	NT06
Severity	High
Priority	High
Bug Title	Online Examination System accepts a submission after the examination deadline.
Environment	Application: Online Examination System; Browser: Google Chrome; Operating System: Windows 10
Preconditions	A student is in an active examination and the configured end time has been reached.
Steps to Reproduce	1. Login. 2. Start scheduled exam. 3. Allow timer to reach zero. 4. Click Submit. 5. Observe submission status.
Expected Result	The system should close or auto-submit according to configured rules and must not accept a late submission.
Actual Result	The system accepts the late submission after the examination time has expired.
Impact	May cause incorrect booking/submission data, financial or academic processing errors, or poor user experience.
Suggested Fix	Validate the condition on both client and server sides and enforce

	the rule again at final submission.
Status	Open — awaiting developer investigation and correction.

7. BUG LIFE CYCLE

New — Tester reports the defect
Assigned — Developer receives the defect
In Progress — Developer investigates and fixes it
Fixed — Developer submits the correction
Retesting — Tester verifies the correction
Closed — Tester confirms the defect is resolved
If the defect still exists after the fix, the tester reopens the bug and sends it back for further correction.